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Cyber Crime Prediction Using the Random Forest Algorithm with Class Imbalance Handling

Abstract-
Currently cybercrime is growing at an alarming pace and it poses a significant problem to the individual, organizations, law enforcement and digital security organizations. Every year, an immense volume of information on the various forms of digital intrusions occurring in various networks is amassed. This data is necessary to analyze in order to detect the latent trends, come up with effective mitigation measures and anticipate future occurrence of similar cyber rime. This analysis can be efficiently performed using big data analytics and machine learning techniques. The given paper proposes a solution to the problem of cybercrime prediction with the use of the Random Forest algorithm, particularly, to the problem of the imbalance between the classes, i.e., the number of malicious events is much smaller than usual

activities. With the use of cybercrime

dataset that contains features like timestamps, IP positions, attack signatures, and protocol types, the study will be capable of predicting the possibility of a particular criminal cyber activity. Although obtaining real-time crime data is a challenging task because of confidentiality guidelines, the current study receives the inputs using the available datasets of the state of art. The given system is confirmed to solve the cybercrime prediction problem, and the experimental findings demonstrate that the implementation of the Random Forest with the imbalance consideration makes better results to find possible solutions and reveal the pattern of crime. This framework helps security professionals to use human and technical resources in the best way possible to prevent any future incidence.

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Keywords: Cybercrime patterns,
Random Forest, Class Imbalance,
Incident-level data, Machine Learning.

I. INTRODUCTION

The cybercrimes are currently occurring at a very high rate posing a significant challenge to law enforcing bodies as well as cyber security agencies across the world. Over the past few decades, much information on different forms of digital intrusions occurring in different parts of the globe has been compiled on a yearly basis. Necessity to study the data collected is indeed very much necessary, such that the possibility to formulate potential ways to solve and minimize cybercrimes and be able to foresee these in the future also becomes available. The issue of the class imbalance essentially challenges the detection of high consequence and rare anomalies, including network intrusions and financial fraud. Regarding the case of cyber security, malicious traffic is typically a very small portion of total traffic and legitimate traffic is typically over 90. This dearth of minority samples makes the model imbalanced, whereby, the algorithms concentrate on the majority sample (legitimate traffic) to reduce the total error, which results in low accuracy of a minority sample (malicious traffic). The traditional means of manual fraud detection and log analysis is not as viable nowadays due to the time-consuming aspects of the traditional means and the inefficiency of the traditional means to process the large volume of the Big Data. This has led to much focus on the machine learning and data mining based computational techniques. Even though they have been applied to logistic regression and SVM classifiers, out of the experiments conducted, it has been discovered that the Random Forest and XGBoost classifiers are quite practical in handling complex structured data. In particular, the Random Forest classifier has been discovered to be less prone to overfitting and it gives a good estimation of the generalization error. The paper will suggest a solution to the prediction problem of cybercrime, namely the Random Forest algorithm and addressing the issue of imbalance of classes. We achieve better prediction of the malicious events by adding data-level interventions, i.e., Synthetic Minority Oversampling Technique (SMOTE), or altering the algorithm, i.e., Weighted Random Forest. On the basis of incident level cybercrime data, which includes data such as timestamps, IP locations, and signature data, we confirm a framework that helps security experts identify trends to assist attacks and security experts assign technical resources to prevent attacks..

II. LITERATURE REVIEW

The cybercrimes are now taking place on an extremely high level which presents a great challenge to both the law enforcing bodies and cyber security agencies in the world. In the last few decades, a lot of data on various types of digital intrusions taking place in various regions of the world has been summarized annually. There is indeed much necessity to study the information gathered, so that the possibility of formulating the potential ways of resolving and minimizing the cybercrimes and be able to anticipate them in future also becomes a possibility. The challenge of the class imbalance is basically the problem of identifying high consequence and rare exceptions, such as network infiltration and financial fraud. In the context of the case of cyber security, malicious traffic is usually very small percentage of total traffic and legitimate traffic is usually more than 90. Such a lack of minority samples makes the model biased, that is, the algorithms focus on the majority sample (legitimate traffic) to minimise the overall error, which leads to low accuracy of a minority one (malicious traffic). The conventional method of manual fraud detection and log analysis is no longer as feasible as it is at present because of the time-consuming nature of the conventional method and the ineffectiveness of the conventional method to handle the high volume of the Big Data. This has given rise to high attention to the machine learning and data mining founded computational methods. Although applied to the logistic regression and SVM classifiers, among experiments performed, it has been found that the Random Forest and XG Boost classifiers are rather practical in the work with the complex structured information. Specifically, the Random Forest classifier has been found to be less susceptible to overfitting and it provides

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